MATHS MANTHAN
• TUITION CLASSES •
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**Subject:** Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 8)** | **Time:** 45 Minutes
**Class:** Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 20**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

## SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

**Q1.** The vertices of a triangle are $O(0,0), A(a,0)$ and $B(0,b)$. The coordinates of its circumcentre are: [1]

(A) $(a,b)$ (B) $\left(\frac{a}{2},\frac{b}{2}\right)$ (C) $\left(\frac{a}{3},\frac{b}{3}\right)$ (D) $\left(\frac{a}{4},\frac{b}{4}\right)$

**Q2.** A circle with centre $P(4,5)$ passes through the point $A(0,9)$. The length of the diagonal of the largest square inside this circle is : [1]

(A) $4\sqrt{2}$ units (B) $8\sqrt{2}$ units (C) $\sqrt{53}$ units (D) $2\sqrt{53}$ units

**Q3.** If the points $A(1,2), O(0,0)$ and $C(a,b)$ are collinear, then: [1]

(A) $a=b$ (B) $a=2b$ (C) $2a=b$ (D) $a=-b$

**Q4.** AOBC is a rectangle whose three vertices are $A(0,2)$, $O(0,0)$ and $B(4,0)$. The square of the length of its diagonal is equal to : [1]

(A) 36 (B) 20 (C) 16 (D) 4

**Q5.** The distance of the point $P(-3,-4)$ from the origin is: [1]

(A) 3 (B) 4 (C) 5 (D) 25

## SECTION B: Short Answer Type Questions (6 Marks)

**Q6.** In what ratio does the point $(2,p)$ divide the line segment joining the points $(-4,2)$ and $(5,-1)$ ? Also, find the value of 'p'. ¿ **OR** ¿ (b) The points $A(3,6)$, $B(k,2)$ and $C(6,2)$ are the vertices of a right triangle ABC right-angled at B. Find the value of k. ¿ ¿ [2]

**Q7.** A point P divides the line segment joining the points $A(-3,5)$ and $B(7,-4)$ in a certain ratio. If the point P lies on the line $y=2x$, then find the ratio $AP:PB$ and coordinates of point P. [2]

**Q8.** Find the coordinates of the midpoint of the line segment joining the points $(2,3)$ and $(4,7)$. [2]

## SECTION C: Long Answer & Competency Based Questions (9 Marks)

**Q9.** Let $D(3,-2), E(-3,1), F(4,-3)$ be mid-points of sides BC, CA, AB respectively of $\Delta ABC$. Find co-ordinates of its vertices A, B and C. **[5]**

**Q10. Competency Case Study:** Trees act the natural filters. By planting trees in and around school premises, we create cleaner and healthier air for students and local residents, reducing respiratory problems. A school in Noida has proposed and organised a community drive on tree plantation under the title "Save Earth, Plant Trees". Students of that school have planted saplings in the field such that it formed a quadrilateral as shown in the figure ABCD. *(From the Cartesian Grid diagram: $A(-5,9)$, $B(2,3)$, $C(1,-5)$, $D(-6,1)$, and intersection of diagonals $P(-2,2)$).* **[4]**

(i) Find the distance between the two saplings at A and D. [1 mark]

(ii) (a) One student plants one sapling at the mid-point of AD. Then he moves along a straight line parallel to DB and sows another sapling on AB. What are the coordinates of the positions of these two new saplings ? [2 marks] — **OR** —

(iii) (b) A new sapling is kept at a point M on DB such that $DM : MB = 3 : 1$. Find the coordinates of M. [2 marks]

(iv) The line segments AC and BD bisect each other at $P(-2,2)$. Find the coordinates of C. [1 mark]

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